DevOps & CI/CD

in Cloud Computing

Principles, pipeline architecture, and practical implementation

CSA1507 - CLOUD COMPUTING FOR BIG DATA ANALYSIS

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What is DevOps?

A culture and set of practices that unify software development (Dev) and IT operations (Ops)

Collaboration

Breaks down silos between developers, testers, and operations teams.

Automation

Automates build, test, and deployment steps to remove manual bottlenecks.

Continuous Improvement

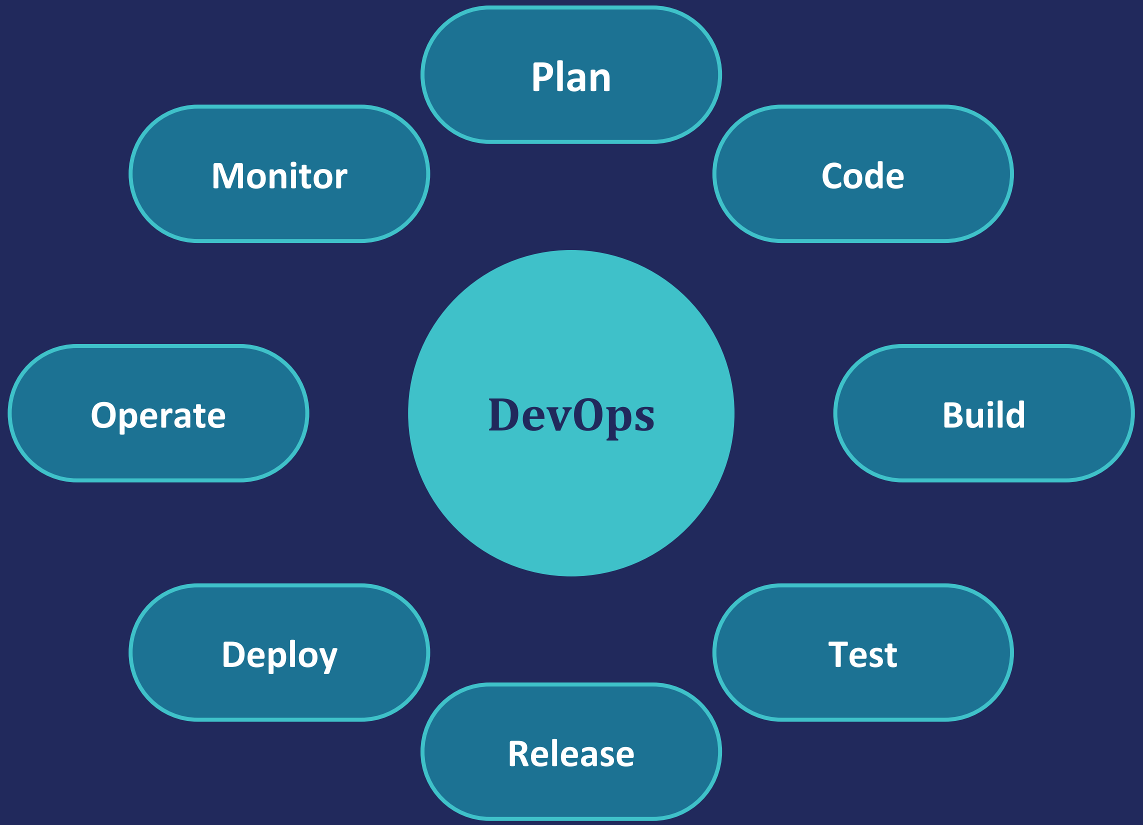
Uses feedback loops and monitoring to iterate rapidly and reliably.

Shared Responsibility

Teams jointly own reliability, security, and speed of delivery.

The DevOps Lifecycle

An infinite loop of planning, building, and operating



What is CI/CD?

Continuous Integration (CI)

- Continuous Integration is the practice of frequently merging code changes into a shared repository, where the code is automatically built and tested.

How it works:

- A developer writes code.
- The code is pushed to a Git repository (such as GitHub).
- A CI tool automatically:
- Builds the application.
- Runs automated tests.
- If all tests pass, the code is ready for the next stage.

Continuous Delivery / Deployment (CD)

- Continuous Delivery ensures that software is always ready to be released. After CI completes successfully, the application is prepared for deployment, but a manual approval is usually required before releasing it to users.

Benefits:

- Faster releases.
- Lower deployment risk.
- Reliable release process.

CI/CD Pipeline Architecture

A typical automated pipeline from commit to production

Source
Code pushed to Git repo

Build
Compile & package app

Test
Unit & integration tests

Package
Create deployable artifact

Deploy
Release to cloud env

Monitor
Observe & alert in prod

Version Control | **Infrastructure as Code** | **Automated Testing** | **Container Orchestration**

Cloud role: cloud platforms (AWS, Azure, GCP) provide elastic build agents, and native pipeline services (CodePipeline, Azure DevOps, Cloud Build) that scale pipelines on demand.

The DevOps Toolchain

Popular tools mapped to each pipeline stage

Source Control	Git, GitHub, GitLab, Bitbucket
CI/CD Engines	Jenkins, GitHub Actions, GitLab CI, CircleCI, Azure Pipelines
Configuration & IaC	Terraform, Ansible, AWS CloudFormation, Pulumi
Containers & Orchestration	Docker, Kubernetes, Helm, Amazon ECS
Monitoring & Logging	Prometheus, Grafana, ELK Stack, Datadog, CloudWatch

Why It Matters: Benefits

10x

More frequent deployments than traditional release cycles

60%

Fewer production failures with automated testing gates

24x

Faster recovery from incidents with CI/CD rollback tooling

- Faster time-to-market for new features and fixes
- Higher software quality via consistent automated testing
- Better collaboration between development and operations
- Scalable, elastic infrastructure aligned with cloud-native design

Practical Example: E-Commerce App on AWS

End-to-end CI/CD implementation using GitHub Actions and AWS

1. Commit & Trigger

Developer pushes code to GitHub; a GitHub Actions workflow triggers automatically on push to main.

2. Build & Test

Workflow installs dependencies, runs unit tests, and builds a Docker image of the application.

3. Push to Registry

Tested image is tagged and pushed to Amazon Elastic Container Registry (ECR).

4. Deploy to Cloud

AWS CodeDeploy (or ECS rolling update) pulls the new image and deploys it to an ECS Fargate cluster.

5. Monitor & Rollback

Amazon CloudWatch tracks health metrics; a failed health check auto-triggers rollback to the last stable version.

Challenges & Best Practices

Common Challenges

- Cultural resistance to cross-team collaboration
- Legacy systems that resist automation
- Security integration ("shift-left") slows initial adoption
- Managing pipeline complexity at scale
- Tool sprawl across teams and environments

Best Practices

- Start small: automate one pipeline stage at a time
- Adopt Infrastructure as Code for repeatable environments
- Bake security scans into the pipeline (DevSecOps)
- Use feature flags for safer, gradual rollouts
- Track DORA metrics to measure real progress

Key Takeaway

DevOps + CI/CD turn cloud infrastructure into a continuous, reliable delivery engine.

Automate relentlessly. Measure everything. Ship with confidence.

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